



Google handwriting recognition

Read the paper that describes how Google uses deep neural networks to recognise handwriting <https://digit.in/sep19-47>



Languages in Gboard

Head over to this list to check out the full list of languages and scripts that are available on Gboard <https://digit.in/sep19-48>

also remains very responsive. If you are putting in a very heavy prediction algorithm, and a lot of machine learning or something like that, then they keypad may not even run on low end phones, or it will be responding very slowly. So Swalekh, right from the start was designed to include all the Indian languages, be absolutely fast and responsive, and install on any device. When we implemented the script grammar itself, we found that most people were able to type everything they needed to, without even using any prediction, because the script grammar was able to predict a lot of things on its own, without requiring a heavy dictionary and all that. Later on, we have added very optimised dictionaries.”

PREDICTING TEXT

The Indus OS team started designing the keyboard by doing extensive research on Indian languages. The team found that as most of the scripts originate from brahmi, there is a correlation in the way the scripts are structured. Indus OS engineers developed algorithms that could take advantage of these similarities between the scripts of all the languages, and also used machine learning to develop features. These efforts resulted in matra predictions, or the symbols representing the vowels, as well as word predictions for the regional languages. These are based on the smartphone understanding what the user is trying to write.

Over time, the engineers have created a corpus that can improve the prediction and consequently, the performance of the keyboard. The corpus contains over two lakh words for each of the languages, and for some languages it even goes up to seven lakh words. This corpus is what makes the predictions so accurate, and almost always the keyboard is capable of anticipating the inputs from the user.

As soon as a user keys in a character, two sets of predictions appear on top. One is the next character or matra, and another is a suggestion for an entire word. Tapping through either of these options makes the typing faster. These features allow any local language user to comfortably type in Indian languages. It also makes it easier for consumers to get used to a new keyboard, as right from the start, the layout is familiar.

At the same time, the engineers accounted for the users who could type

using an English language keyboard, or found it more comfortable. For this reason, it is also possible to type in indic scripts, using an english keyboard. Depending on the kind of text entered, users have the option to either transliterate or translate the entered text. These features are realised through a simple swipe mechanism, swipe left to transliterate, and swipe right to translate. The keyboard is in fact named after this innovation. Us-

ing the translation option allows users who do not know a writing in a native language to easily communicate with users who know to read exclusively in a native language.

Even for Gboard, prediction and machine learning plays an important role in improving the typing experience. Daan explains, “Each language and script that we work on poses unique challenges. Every time a new language is added to Gboard, we have to decide where the keys go on the layout (for example, do we use a QWERTY layout or an AZERTY layout). We also create a new machine learning language model for each language. This model trains Gboard to know when and how to autocorrect your typing, or to predict your next word. In order to train our machine learning language models, for each language that we want to add, we need a text

corpus (which is a database of lots of available texts that are written in a particular language).”

One of the problems with creating such a corpus is the availability of resources in the local languages, such as digitised books, news articles or court documents. Some of the languages supported by Gboard such as Awadhi and Bhili are primarily spoken. Daan says, “for languages like English, which has only about 30 characters and large amounts of written materials widely available, designing a layout and getting text corpora was easy. But, in the case of Indian languages, this process is much harder. Many Indian languages have a small presence on the web, so we worked closely with native speakers across India to collect data to build text corpora so we can train our machine learning models, helping people to start communicating easily in their native script on their own smartphone.”

The transliteration method of input relies on machine learning as well. “Layout design is also an interesting challenge for many Indian languages, since there are lots of different characters that need to be typed. In Gboard, we designed “dynamic” layouts, where you can type a consonant, and the different vowel options appear at the top of the layout. But another very popular input method for Indian languages, and other languages with complex scripts, is to use transliteration, romanizing the native-script characters into the Latin alphabet, often according to their phonetic pronunciations. We use machine learning to let users type in romanized Indian languages, turning the output back into the native script. For example, our transliterated Hindi keyboard allows users to type “daanth” and get “दांत” (teeth).”

As the keyboard is dynamic, the sequence of keys and the content of the message dictates the layout, which is constantly changing according to the current needs of the user. The keys that keep changing are indicated by blue colouring, while all the other keys remain black. Daan says, “Layout



The Indus Swipe Keyboard